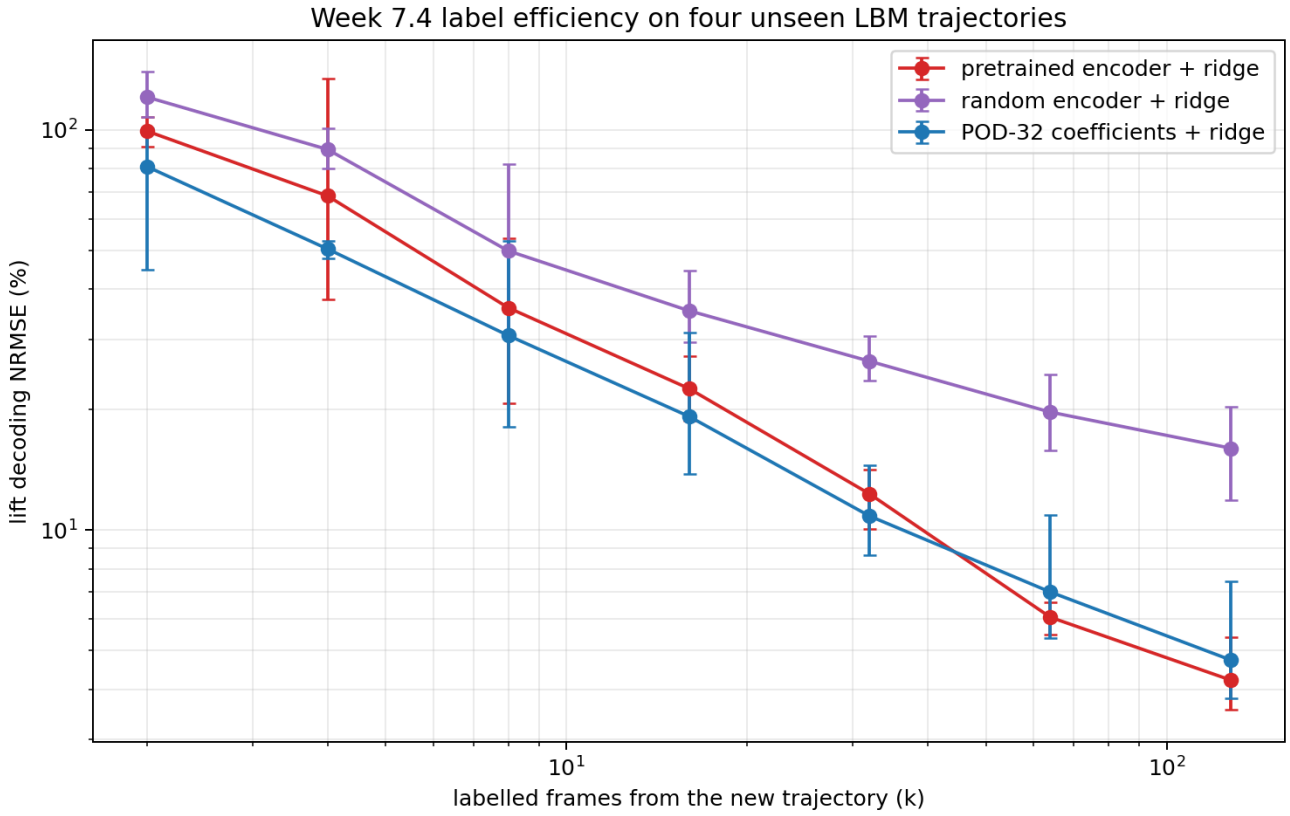
Week 7.4

Diverse-wake pretraining and label-efficient force decoding

A trajectory-split, MAPA-inspired experiment on 16 educational LBM wakes



Why a new week?

Week 7.3 asks a network to complete a periodic low-rank field, a task gappy POD almost solves analytically. MAPA transfers a representation learned from heterogeneous unlabelled signals to an external label. Week 7.4 follows that design: the MAE sees velocity only; the downstream target is cylinder lift.

Frozen split

Development: Re 60, 65, 70, 75, 80, 85, 90, 100, 110, 120, 130. Validation: Re 105. Untouched test: Re 95, 115, 125, 135. Target labels come from frames [0,150); frames [180,251) are scored. No random frame split crosses trajectory roles.

Model

A 90k-parameter masked autoencoder hides 75% of 4 by 6 patches and trains for 6000 AdamW steps. The encoder sees visible tokens only. Downstream, all visible tokens are mean pooled to a 64-component representation; ridge strength is selected on Re105.

Baselines

The identical random encoder tests the value of pretraining. POD-32 coefficients provide a strong classical representation. Every representation receives exactly the same k labels and ridge protocol.

Definitions and reading the metric

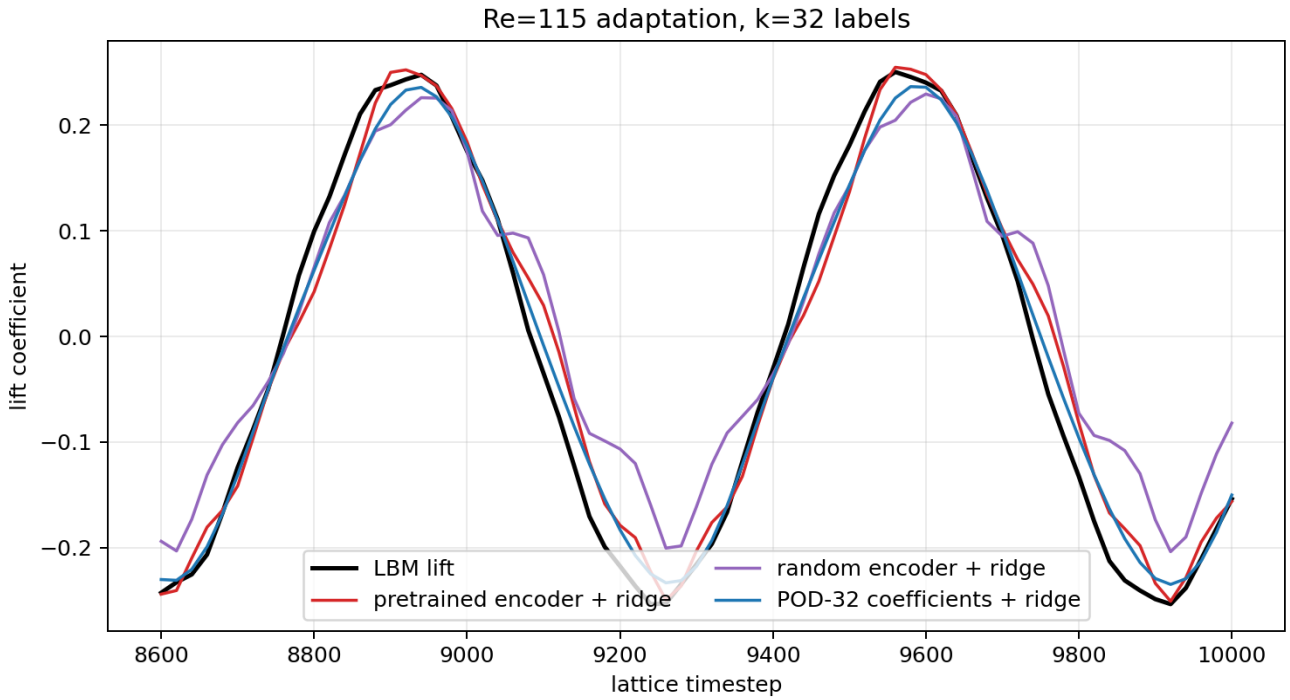
A 32 by 78 snapshot is divided into 104 patches, each with 24 velocity values. The hidden-patch MSE supplies the pretraining signal. The downstream ridge readout minimizes squared lift error plus an L2 penalty on its weights. NRMSE (%) is 100 times RMSE divided by the scored lift standard deviation. The current field is available when decoding current lift; this is not an autonomous forecast. Lift is produced alongside the solver fields, so label scarcity is simulated rather than an observed saving in CFD cost.

Retained numerical evidence

Lift NRMSE (%), mean across four unseen trajectories

k	Pretrained	Random	POD-32
2	99.44	120.91	81.06
4	68.46	89.45	50.44
8	35.96	49.89	30.65
16	22.59	35.35	19.24
32	12.31	26.43	10.84
64	6.05	19.72	7.00
128	4.22	16.02	4.73

At k=32, pretrained 12.31% is below random-at-k=128 16.02%: better mean accuracy with one-quarter as many target labels. POD leads through k=32; pretrained passes it in the mean at k=64 and 128.



Scope and reproducibility

Ridge selection uses 2761 development and 251 validation lift labels in addition to the reported target-label budget. Only one encoder initialization is retained; the three seeds are label draws. POD rank 32 is fixed, not rank-optimized. One run per Reynolds number confounds Reynolds and initial-condition variation. The test cases are now inspected retained evidence; future tuning needs fresh tests. Mean differences are not statistical significance claims.

The retained success is representation transfer for an external force label, not a claim that transformers universally beat POD. The solver uses D=6 quick D2Q9-TRT trajectories and periodic transverse boundaries; the data are not grid-independent DNS. Re140 was rejected after the solver became non-physical. The histories do not support a validated Strouhal claim.

Execution: the notebook verifies the 37.1 MB release before optional training. RUN_FULL uses a fresh output directory. The runner also defaults to a temporary directory and refuses retained evidence paths. For data regeneration use qa/generate_week07_4_wakes.py; inspect hashes and every retained record in results/week07_4_diverse_pretraining/metrics.json.

References

Tang, Spalding and Cogan (2026), Pretraining for Sample-Efficient Neural Interfaces (MAPA), arXiv:2609.13507. He et al. (2022), Masked Autoencoders Are Scalable Vision Learners. Everson and Sirovich (1995), Karhunen-Loeve procedure for gappy data.